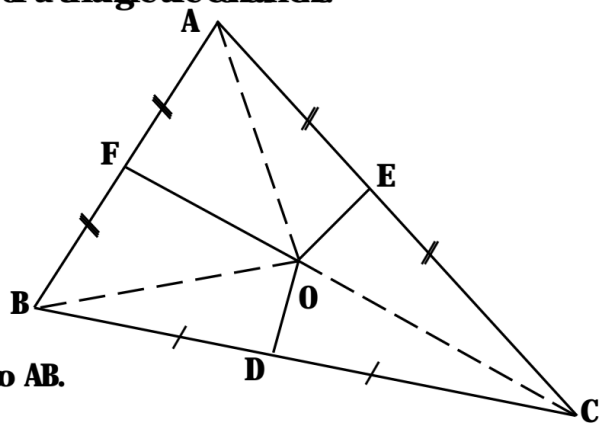
11. Concurrencies connected with a Triangle

1. Perpendicular bisectors of the sides

The perpendicular bisectors of the three sides of a triangle are concurrent.



Given : OD and OE are the perpendicular bisectors of the sides BC and CA respectively of a triangle ABC. They intersect at O. Let F be the mid point of BA and join OF.

To prove : OF is perpendicular to AB.

Construction : Join OA, OB, OC.

Proof : In triangles BDO, CDO

BD = CD (given)

OD = OD (common)

$\angle BDO = \angle CDO$ **(=90^0**, given)

$\Delta BDO \equiv \Delta CDO$ (SAS)

Therefore OB = OC ----------(1)

Similarly $\Delta OCE \equiv \Delta OAE$

Therefore OC = OA ----------(2)

From (1) and (2) OA = OB.

In triangles OAF, OBF

OA = OB (proved)

OF = OF (common)

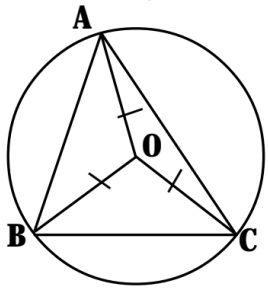
AF = FB (given)

$\Delta OAF \equiv \Delta OBF$ (SSS)

$\angle OFA = \angle OFB = 90^0$

Therefore OF is perpendicular to AB.

Hence, the three perpendicular bisects of the sides of a triangle meet at a point.

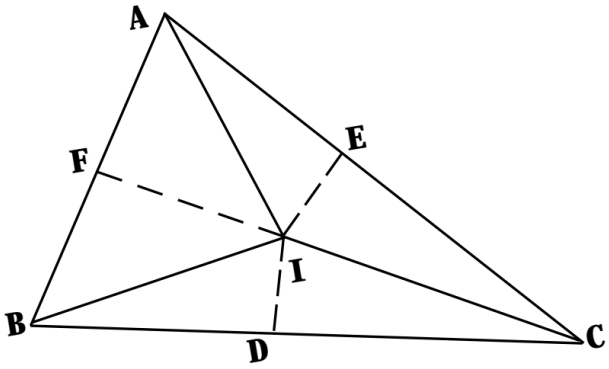


In this diagram, OA=OB=OC. O is said to be the *circumcentre* **of the triangle ABC. The circle is called** *circumcircle* **of the triangle ABC.**

2. Bisectors of the angles

The bisectors of the three angles of a triangle are concurrent.

Given : **IB and IC are the bisectors of the angles ABC and ACB of a triangle ABC. Join AI.**



To prove : **AI bisects the angle BAC.**

Construction : **Draw perpendiculars ID, IE and IF to BC, CA and AB respectively.**

Proof : **In triangles BDI, BFI**

$\angle IBD = \angle IBF$ **(given)**

$\angle BDI = \angle BFI$ **(=90**^{0}, given)

BI = BI (common)

$\Delta BDI \equiv \Delta BFI$ (RHS)

$\therefore ID = IF$ ——————— **(1)**

Similarly it may be proved that

$\Delta CDI \equiv \Delta CEI$

$\therefore ID = IE$ ——————— **(2)**

From (1) and (2), IE = IF.

In triangles AEI, AFI

IE = IF (proved)

IA = IA (common)

$\angle IEA = \angle IFA$ (=90^0)

$\Delta AEI \equiv \Delta AFI$

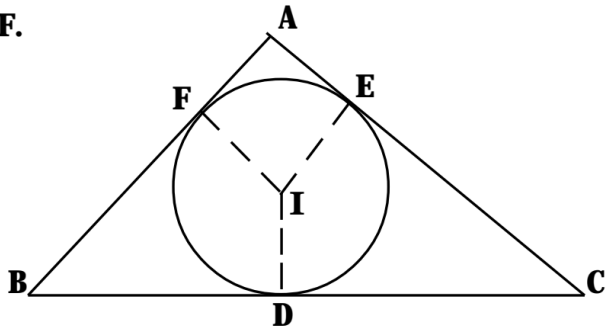
Therefore $\angle EAI = \angle FAI$

ie. IA is the bisector of the angle BAC.

Hence, the bisectors of three angles of a triangle are concurrent.

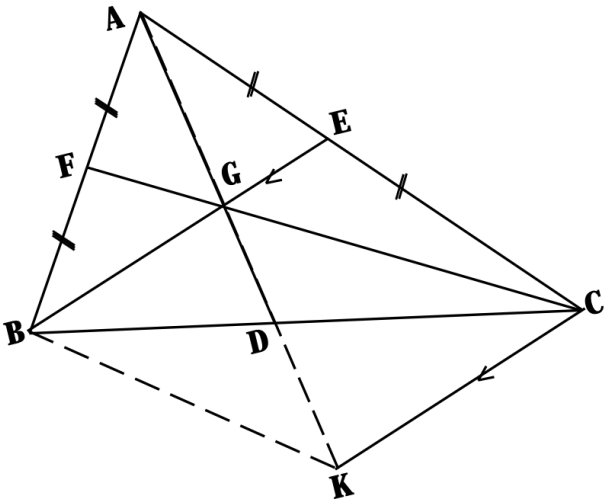
It is said to be *incentre* **and the circle is called** *inscribed circle.*

radius = ID = IE = IF.



3. Medians

The three medians of a triangle are concurrent.



Given : E and F are the mid points of the sides AC, AB of a triangle ABC. BE and CF meets at G. Join A G and produce it to meet BC at D.

To prove: BD = DC

Construction : Through C draw CK parallel to BE; produce AD to meet CK at K. Join BK.

Proof : In the triangle AKC,

AE = EC (given)

EG // CK (construction)

Therefore, AG = GK ——————— (1)

In the triangle A B K

AF = FB (given)

AG = GK (proved)

Therefore, FG // BK ——————— (2)

In the quadrilateral CGBK

CK // GB

BK // GC

Thefefore CGBK is a parallelogram

The diagonals of a parallelogram bisect each other.

BD = DC and GD = DK

BD = DC means AD is a median of the triangle ABC.

Hence the three medians of a triangle meet at a point.

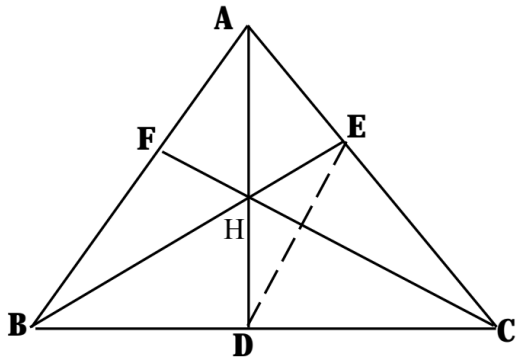
The point (G) of intersection of three medians is called *centroid* of the triangle.

4. Altitudes

The perpendiculars drawn from the vertices of a triangle to the opposite sides are concurrent.

Given : **Let AD and BE be the perpendiculars drawn from A and B to the opposite sides and let them intersect at H**
Join CH and produce it to meet AB at F.

To prove : **CF is perpendicular to AB.**



Construction : **Join DE.**

Proof : **In the quadrilateral DCEH**

$\angle CDH + \angle CEH = 90^0 + 90^0 = 180^0$

DCEH is a cyclic quadrilateral.

$\angle DEC = \angle DHC$ **(angle in the same segment)** ———————— **(1)**

$\angle DHC = \angle AHF$ **(the vertically opposite angles)**

since $\angle AEB = \angle ADB$ **(=90^0)**

AEDB is cyclic quadrilateral.

$\angle ABD = \angle DEC$ (exterior angle = interior opposite angle)——(2)

From (1) and (2) $\angle ABD = \angle DHC$

Therefore, BDHF is a cyclic quadrilateral.

$\angle HDB + \angle HFB = 180^0$

$\angle HFB = 90^0$

i.e., CF is perpendicular to AB.

Hence the three perpendiculars AD, BE and CF meet at H.
H is called ***orthocentre***.

Exercise

1 **The sides AB, AC of a triangle ABC are produced. Show that the bisectors of the exterior angles of B and C and the bisector of the interior angle of A are concurrent.**